

The empirical recurrence for OEIS A229684, recovered from its tail

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Abstract

OEIS A229684 records “Empirical recurrence of order 46” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 132$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 48$. Its last coefficient is $c_{46} = -8862938119652501095929000000000000 \neq 0$, so no further tail satisfies anything shorter; any order-46 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229684 is “Number of defective 3-colorings of an $n \times 7$ array connected diagonally and antidiagonally with exactly two mistakes, and colors introduced in row-major order”. It has offset 1 and begins

0, 101376, 7952256, 437924880, 21063718224, 937213830720, 39647663129952, 1617006498774912, ...

The entry states:

Empirical recurrence of order 46 (see link above).

As of the “Last modified” line on the live entry (Apr 28 2021 01:22:28, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 132$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 132$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 46: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 46.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 264$ exact terms from $n = 2$ on, it returns order 46 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{46} c_i a(n-i),$$

c_1	120	c_{13}	−82020735657038736	c_{25}	2376148218576349173139245312
c_2	−4563	c_{14}	2124541854056257272	c_{26}	−8652712634186054877183620880
c_3	4320	c_{15}	−474964638098416704	c_{27}	−29945136514920863746825445760
c_4	3784158	c_{16}	−175170019654805362812	c_{28}	197033402367240698142156902400
c_5	−66945528	c_{17}	626107210532361851904	c_{29}	124583238616546145129110560000
c_6	−942029838	c_{18}	9075926345894285043852	c_{30}	−2740314086124042343480621920000
c_7	35637794856	c_{19}	−61244316640103403969504	c_{31}	2530513171146741422421355776000
c_8	−25501668777	c_{20}	−267700910858345844103536	c_{32}	23674706929335713543142959160000
c_9	−8867171659536	c_{21}	3278146062315936924166464	c_{33}	−52303433270908488143058270720000
c_{10}	63733879930797	c_{22}	1876002849109756416573360	c_{34}	−113734090284218497589183818200000
c_{11}	1211496571856376	c_{23}	−110530778690244699555859584	c_{35}	471488383322942360501707588800000
c_{12}	−15940067071492872	c_{24}	186344770852691314134751440	c_{36}	118654808554390277597222268000000

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 48$, and it is the recurrence OEIS A229684 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{46} - c_1 x^{45} - \dots - c_{46}$. Its constant term is $-c_{46} = 8862938119652501095929000000000000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 46 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 46, so $\deg q = 46$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 46, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 46, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is $-8862938119652501095929000000000000$.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-46 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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